

Lesson 1: Classification of Engineering Materials

Learning objectives

- Name the four main classes of engineering materials.
- Give two examples and two uses for each class.
- Explain why materials are chosen for particular jobs.

1.1 Why classify materials?

Engineers choose a material by looking at what the product must do, how long it must last and how much it may cost. Grouping materials into classes makes this choice easier because materials in the same class behave in similar ways.

1.2 The four main classes

Metals are strong, conduct heat and electricity and can be shaped by casting, forging or machining. They are divided into ferrous metals (contain iron, such as steel and cast iron) and non-ferrous metals (such as copper, aluminium and brass).

Polymers (plastics and rubbers) are light, cheap and resist many chemicals. Thermoplastics soften when heated and can be re-moulded, while thermosetting plastics harden permanently.

Ceramics such as porcelain, glass and bricks are hard and heat resistant but brittle.

Composites combine two or more materials to get better properties, for example reinforced concrete and fibreglass.

Class	Examples	Main properties	Common uses
Metals	Steel, copper, aluminium	Strong, conductive, ductile	Bridges, wires, engines
Polymers	PVC, polythene, rubber	Light, insulating, corrosion free	Pipes, bags, tyres
Ceramics	Glass, porcelain, brick	Hard, brittle, heat resistant	Tiles, insulators, cookware
Composites	Concrete, fibreglass	Strong and light for their size	Buildings, boats, helmets

Key points

- Metals are ferrous or non-ferrous.
- Thermoplastics can be re-melted; thermosets cannot.
- Composites combine materials to gain the best of each.

Review questions

- 1 List the four classes of engineering materials.
- 2 Why is copper used for electrical wires?
- 3 State one difference between a thermoplastic and a thermosetting plastic.

Lesson 2: Mechanical Properties of Materials

Learning objectives

- Define strength, hardness, ductility, toughness and elasticity.
- Calculate simple stress values.
- Match properties to everyday uses.

2.1 What are mechanical properties?

Mechanical properties describe how a material responds when a force is applied. They decide whether a material will bend, stretch, snap or wear away in service.

2.2 Key properties

Strength is the ability to withstand a load without breaking.

Hardness is resistance to scratching, denting or wear.

Ductility is the ability to be drawn into a wire, and **malleability** is the ability to be hammered into thin sheets.

Toughness is the ability to absorb energy and resist cracking under impact.

Elasticity is the ability to return to the original shape after the load is removed.

2.3 Simple stress calculation

Stress is the force acting on each unit of area: **Stress = Force / Area**.

Worked example: A steel rod with a cross-section of 200 mm^2 carries a pull of 10 000 N. Stress = $10\,000 / 200 = 50 \text{ N/mm}^2$ (50 MPa).

Property	Meaning	Everyday example
Hardness	Resists scratching and wear	Drill bits, file teeth
Ductility	Can be drawn into wire	Copper wire
Toughness	Absorbs impact without cracking	Hammer head
Elasticity	Springs back after loading	Steel spring

Key points

- Stress = Force / Area.
- Ductile materials stretch before they break; brittle materials snap suddenly.
- Tough is not the same as hard.

Review questions

- 1 Define ductility and give one example of a ductile metal.
- 2 A force of 6 000 N acts on an area of 300 mm^2 . Calculate the stress.
- 3 Why is glass considered brittle?

Lesson 3: Corrosion and Its Prevention

Learning objectives

- Explain what corrosion is and why it happens.
- Describe three methods of preventing corrosion.

3.1 What is corrosion?

Corrosion is the gradual destruction of a metal by chemical reaction with its surroundings. The rusting of iron needs both oxygen and water, and it is faster in salty coastal air.

3.2 Preventing corrosion

Barrier coating: paint, oil, grease or plastic coating keeps air and water away from the metal.

Galvanising: steel is coated with a layer of zinc. Even if the coating is scratched, the zinc corrodes first and protects the steel.

Alloying: stainless steel contains chromium, which forms a thin protective oxide layer.

Cathodic protection: a more reactive metal block (a sacrificial anode) is attached to the structure, such as a ship hull, so that it corrodes instead.

Method	How it works	Typical use
Painting	Blocks air and water	Steel roofs, vehicles
Galvanising	Zinc layer protects steel	Roofing sheets, fences
Stainless steel	Chromium oxide film	Kitchen sinks, cutlery
Sacrificial anode	Reactive metal corrodes first	Ship hulls, pipelines

Key points

- Rusting needs oxygen and water.
- Coastal and humid climates speed up corrosion.
- Sacrificial protection uses a more reactive metal.

Review questions

- 1 Which two substances are needed for iron to rust?
- 2 Explain how galvanising protects steel.
- 3 Suggest two ways to protect a steel gate kept outdoors.